

F1 for speed

There's more to F1 than laps, speed, burning rubber and champagne. Read to know the technology behind designing an F1 car

Green Heights

Taipei 101, the second tallest building in the world, expected to be the tallest Green Building in the world in 18 months

Reaching out to the skies

Nishith Rastogi

readersletters@thinkdigit.com

Did you know the world's tallest building, the *Burj Khalifa*, and mobile phone company Samsung have a connection?

Skyscrapers such as the *Burj* have come a long way from The Home Insurance Building, constructed in 1885 and commonly referred as the first skyscraper, which was merely 138 feet in height and had ten stories. Today, Burj Khalifa is the tallest freestanding structure made by man with a staggering height of 828 metres. If the name Burj Khalifa sounds unfamiliar, that's because Burj Dubai at the last moment was renamed to Burj Khalifa in the honour of the ruling president of Abu Dhabi, Sheikh Khalifa bin Zayed al Nahyan who provided a \$10 billion bailout for the financially troubled Dubai government.

There's no exact definition to what qualifies as a skyscraper and is more like a variable with time, and geographic region as it's dependent parameter. Currently, in the US as a matter of convention, buildings greater than 150 metres (approximately 45 stories) are classified as skyscrapers. As you'd know, a building in India with a height of over 100 metres would be clearly distinguishable from its surroundings and would earn the title of being a skyscraper.

The ability of living and working 100 stories above the ground present its own set of unique challenges ranging in domain from construction to plumbing. Here's a look at some of the problems faced by designers and civil engineers while constructing a skyscraper and their solutions.

Construction

Given their height, skyscrapers either sink in to the ground, or collapse due to

instability, and hence they are built around a central frame generally made of steel. Think of the steel frame as a three dimensional lattice of interconnected steel bars on which the weight of the whole building rests. This eliminates the need of the load bearing shear walls, and instead can be replaced by curtain wall generally made of glass giving massive weight advantages. This is the main reason to why skyscrapers have *glass* walls.

Most modern (post 1970) skyscrapers use the "Tube Frame" which was introduced by Fazlur Rahman Khan. This was done to resist lateral forces of wind and storm. Here, the whole building acts like one hollow beam positioned perpendicular to the ground surface. The exterior of the building is reinforced with intersecting spandrel beams.

A major challenge in building a super-heavy skyscraper is creating a stable foundation for the building. For example, consider the Burj Al Arab, famously known as the yacht shaped hotel in Dubai. It took close to three years to reclaim the land from the sea and form the foundation. The rest of the building was completed in approximately the same time frame. To maintain stability, 230 concrete pipes (each 40-metre long) were drilled in to the ground.

Similarly, the Taipei 101 tower built in earthquake prone Taiwan has 382 of such concrete pipes (each of which is 30 metres in height and 1.5 metres in diameter) drilled into the ground to secure the building from earthquake. Approximately 15 months were devoted to the construction of this foundation. It was the tallest building at the time of its construction.

In the case of Burj Khalifa, there were 192 pipes with lengths exceeding 50 metres each. Reinforced concrete forms the main structural system for the building, a total of 3,33,000 cubic metres

